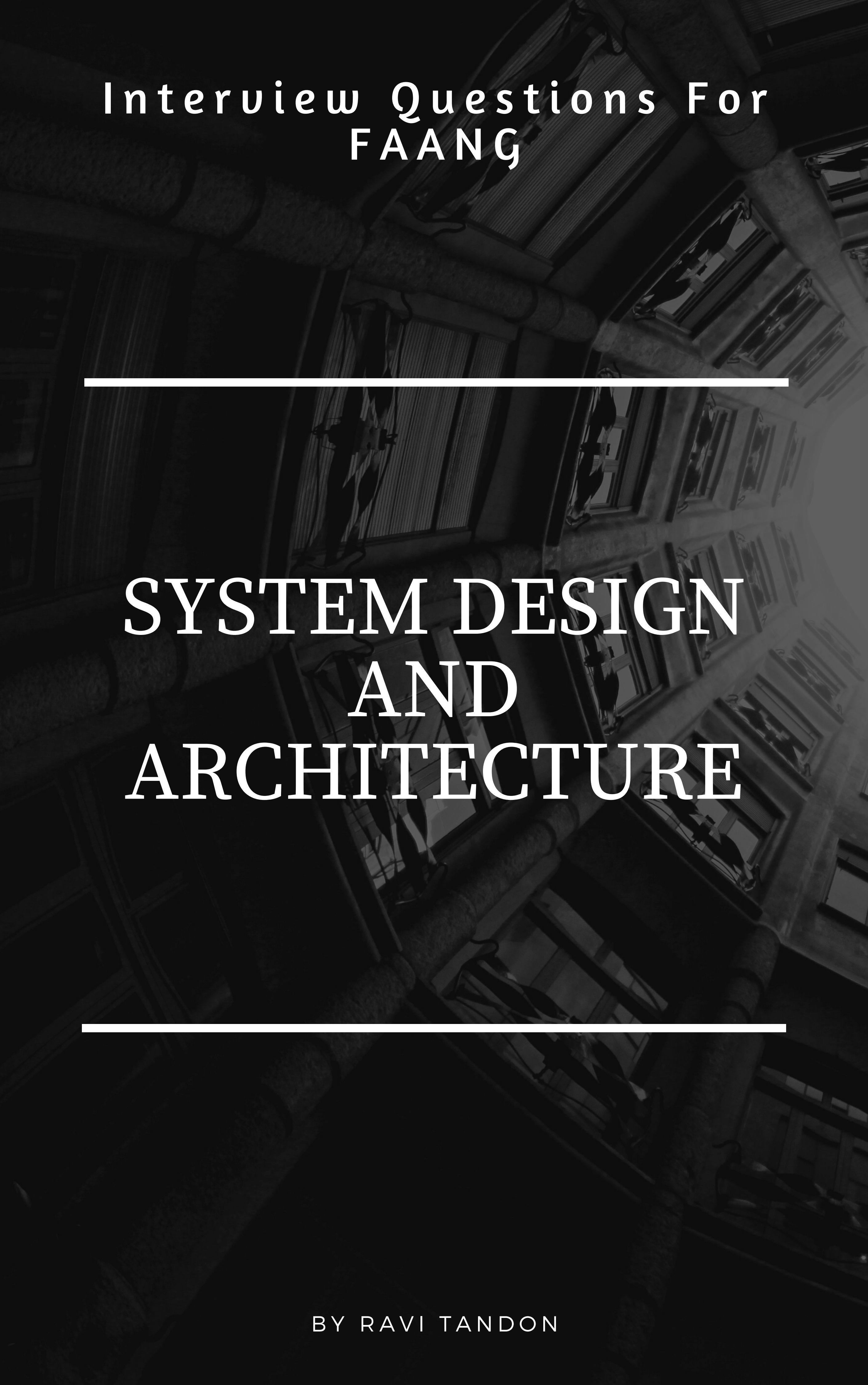




Interview Questions For  
FAANG

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SYSTEM DESIGN  
AND  
ARCHITECTURE

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BY RAVI TANDON

# Chapter 1: The Fundamentals Of System Design

# The Elements Of A Distributed Systems

A distributed system is defined as “An interconnected set of nodes that are linked by a network and share information between them.” Although a distributed system consists of multiple components, it serves a single purpose to the system’s user. Some examples of distributed systems in computer science are the backends of Google, Facebook, LinkedIn, etc. Systems like the Internet, Blockchain or even the human nervous system is a distributed system. For the sake of this blog series, we will focus on only those in Computer Systems 😊😊

We will discuss some of the critical elements of a typical distributed system for building user-facing applications next.

## Microservices

The microservice architecture is a design paradigm where an application is structured to collect several independent services. The characteristics of these services are as follows:

- Organized around a specific business capability
- Own by a small team
- Independently deployable
- Highly scalable
- Loosely coupled
- Highly maintainable and testable

## Load Balancers

A load balancer is a device that acts as a reverse proxy and distributes the network traffic across several different servers. Load